Reproducible 6-DOF Wrist-Roll Retrofit for OpenMANIPULATOR-X

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***Abstract*—Low-cost educational manipulators lack the full orientation control required for complex manipulation tasks due to their limited degrees of freedom. This paper presents a complete and reproducible retrofit to upgrade the OpenMANIPULATOR-X from a 5-DOF [1] to a 6-DOF [2] system by integrating [3] an additional wrist-roll joint [4], accompanied by a unified hardware and ROS 2–based software stack.**

The proposed approach includes a detailed mechanical retrofit kit (CAD models, bill of materials, and assembly procedure), an updated URDF/Xacro robot description with calibrated joint limits and dynamics, and a full ROS 2 bring-up incorporating ros2_control, controllers, and MoveIt 2 motion planning. A structured benchmarking framework is developed to systematically compare the 5-DOF and 6-DOF configurations under identical conditions, evaluating kinematic feasibility, orientation reachability, motion efficiency, and planning robustness in constrained environments.

Experimental results demonstrate that the upgraded 6-DOF system achieves complete orientation reachability across all tested configurations, whereas the baseline 5-DOF system succeeds only in limited cases. In constrained planning scenarios, the 6-DOF configuration achieves a 25.0% success rate compared to 0.0% for the 5-DOF system, with statistically significant improvement ($\chi^2 = 18.81$, $p < 0.000025$). Additionally, the 6-DOF system reduces joint-space motion by 25.8% for equivalent tasks and maintains high control accuracy, achieving sub-degree orientation error (0.55) and consistent trajectory tracking across repeated trials.

These results demonstrate that the addition of a wrist-roll joint significantly enhances kinematic capability, planning success, and execution efficiency while preserving precision and system stability. The complete set of open-source artifacts, including hardware designs, ROS 2 packages, and benchmarking tools, establishes a practical and accessible upgrade pathway for educational manipulators, enabling more advanced robotics research, reproducible experimentation, and broader adoption in academic laboratories.

I. Introduction

Low-cost educational manipulators are widely used in robotics laboratories for instruction, rapid prototyping, and benchmarking. Platforms such as the OpenMANIPULATOR-X provide accessible integration with ROS 2 and MoveIt 2, enabling exploration of perception, motion planning, and control within compact manipulation systems. However, such platforms are typically limited in their degrees of freedom, constraining end-effector orientation capability.

In the standard 5-DOF configuration, the absence of an independent wrist-roll joint restricts the manipulator's ability to achieve arbitrary orientations at a fixed spatial position. This limitation becomes critical in tasks requiring precise alignment, insertion, or orientation-constrained manipulation, where feasible inverse kinematics solutions may not exist or require compensatory joint motions. These compensations degrade motion efficiency, increase execution time, and reduce planning success rates. Despite the widespread adoption of 5-DOF educational manipulators, the quantitative impact of this limitation has not been systematically characterized under controlled and reproducible conditions.

Prior work has explored mechanical extensions of low-cost manipulators; however, many approaches lack either a fully reproducible software stack or a rigorous, statistically grounded comparison between baseline and modified configurations. Consequently, the practical benefits of increased kinematic redundancy remain insufficiently validated.

This work presents a fully reproducible 6-DOF wrist-roll retrofit for the OpenMANIPULATOR-X, extending the original architecture through mechanical integration and updates to the robot model, control stack, and motion planning pipeline using ROS 2, ros2_control, and MoveIt 2. The design preserves compatibility with the baseline platform while enabling independent end-effector roll control.

To evaluate the impact of the additional degree of freedom, a structured benchmarking framework is developed to compare 5-DOF and 6-DOF configurations under identical planning conditions. The evaluation considers kinematic feasibility, planning robustness, and motion efficiency within a shared workspace. Hardware experiments further validate controller performance, trajectory tracking accuracy, and end-effector repeatability, while additional sim-to-real analysis quantifies consistency between simulation and physical execution.

Contributions

The main contributions of this work are as follows:

- A validated 6-DOF wrist-roll mechanical retrofit for the OpenMANIPULATOR-X.
- An updated URDF/Xacro model with ROS 2 control integration for both simulation and hardware.

- A complete MoveIt 2 configuration for motion planning with the extended kinematic chain.
- A reproducible benchmarking framework comparing 5-DOF and 6-DOF configurations.
- A comprehensive evaluation demonstrating improvements in kinematic capability, planning performance, and physical system reliability.

The remainder of this paper is organized as follows. Section II reviews related work. Sections III and IV describe the hardware design and software implementation, respectively. Section V presents the benchmarking methodology, followed by the ablation study in Section VI. Experimental results are reported in Section VII and discussed in Section VIII. Finally, Section IX concludes the paper and outlines future research directions.

II. RELATED WORK

Low-cost robotic manipulators have become essential tools in robotics education and academic research due to their affordability, modularity, and compatibility with modern software ecosystems. Platforms such as the OpenMANIPULATOR-X , WidowX , and Franka Emika Panda span a range of cost and capability, enabling applications in manipulation research, imitation learning, and curriculum-driven instruction . Among these, the OpenMANIPULATOR-X is widely adopted for its Dynamixel-based architecture and seamless integration with ROS 2 and motion planning frameworks. However, its standard 5-DOF configuration imposes inherent kinematic limitations, particularly in tasks requiring precise end-effector orientation control .

The relationship between manipulator degrees of freedom and task-level capability is well established. A minimum of six DOF is required to achieve arbitrary end-effector poses in three-dimensional space . Manipulators with fewer DOF, such as 5-DOF systems, are restricted to a subset of achievable orientations at a given position, resulting in coupling between position and orientation. Classical robotics analyses, including those by Craig and Whitney , highlight that the absence of an independent wrist-roll joint prevents full specification of orientation, leading to infeasible inverse kinematics solutions or inefficient compensatory motions. While these limitations are theoretically understood, they are often discussed qualitatively and lack systematic, experimentally validated characterization on widely used low-cost platforms.

Modern robotic manipulation systems are typically built upon the Robot Operating System 2 (ROS 2), which provides a flexible middleware for communication and control. The ros2_control framework offers a standardized hardware abstraction layer, while MoveIt 2 enables motion planning, kinematic modeling, and collision-aware trajectory generation. Underlying these systems, the Open Motion Planning Library (OMPL) provides a suite of sampling-based planners, including RRTConnect, PRM, and EST, which are widely used for high-dimensional motion planning problems. Benchmarking tools such as MoveIt's evaluation suite and Planner Arena have further enabled systematic comparison of planning algorithms using metrics such as success rate, computation time, and path quality.

Motion planning in constrained environments introduces additional challenges, particularly when orientation constraints must be maintained within narrow workspaces. Prior work by Berenson *et al.* introduced constraint-based planning approaches to address such scenarios, emphasizing the importance of kinematic feasibility under geometric restrictions. These methods highlight the critical role of manipulator dexterity in achieving feasible solutions in tightly constrained environments, motivating the need for additional degrees of freedom in such contexts.

Parallel to these developments, reproducibility has emerged as a key concern in robotics research. Efforts such as standardized benchmarking frameworks and open evaluation platforms, including the Benchmark for Robot Learning, emphasize the importance of sharing complete experimental artifacts and protocols to enable consistent and verifiable comparisons. Despite these advances, many prior works on extending low-cost manipulators focus primarily on hardware augmentation without providing a fully integrated software pipeline or statistically rigorous evaluation. As a result, the practical benefits of added kinematic redundancy, such as a wrist-roll joint, remain insufficiently quantified under controlled and reproducible conditions.

In summary, although low-cost manipulators, advanced motion planning frameworks, and benchmarking tools are widely available, there remains a lack of unified studies that combine mechanical augmentation, full software integration, and rigorous evaluation. In particular, the impact of introducing an additional wrist-roll degree of freedom on kinematic feasibility, planning robustness, and execution efficiency has not been systematically evaluated on commonly used platforms.

To address this gap, this work presents a fully reproducible 6-DOF wrist-roll retrofit for the OpenMANIPULATOR-X, integrating mechanical modification with a complete ROS 2 and MoveIt 2 pipeline. A structured and statistically grounded benchmarking framework is developed to enable direct comparison between 5-DOF and 6-DOF configurations under identical conditions, providing quantitative insights into the benefits of enhanced kinematic capability.

III. HARDWARE DESIGN

A. Design Philosophy

The retrofit is guided by three principles: mechanical minimalism (disturb as few existing components as possible), full reversibility (the original 5-DOF arm can be restored by removing the added motor and restoring the original `link5.stl` mesh), and low incremental cost (leverage existing Dynamixel infrastructure and 3D-printable parts). Adding

TABLE I
DYNAMIXEL MOTOR CONFIGURATION AFTER RETROFIT

ID	Joint	Mode	Status
11	joint1 (base)	Position (3)	unchanged
12	joint2 (shoulder)	Position (3)	unchanged
13	joint3 (elbow)	Position (3)	unchanged
14	joint4 (wrist pitch)	Position (3)	unchanged
15	joint5_roll	Position (3)	**new**
16	gripper_left_joint	Curr-pos. (5)	**moved from 15**

a single additional revolute joint between the existing wrist-pitch link (*link5*) and the gripper assembly achieves a true sixth degree of freedom—independent end-effector roll about the approach axis—without redesigning the arm structure.

B. Motor Selection and Physical Configuration

The new wrist-roll joint is actuated by a Dynamixel XL430-W250-T servo, selected for compatibility with the existing RS-485 daisy-chain bus and U2D2 interface, its compact form factor (approximately 33.5 mm axial depth), and its ability to operate in Position Control Mode (Mode 3), consistent with the other arm joints. The motor is assigned Dynamixel ID 15; the existing gripper servo, previously ID 15, is reassigned to ID 16 to avoid address conflicts. This reassignment must be performed using Dynamixel Wizard before powering the full system. Table I lists the complete motor configuration after the retrofit.

C. Mechanical Components — Retrofit BOM

Table II lists *only the additional components* required beyond the standard OpenMANIPULATOR-X assembly. The base arm, its four Dynamixel servos, gripper, power supply, and U2D2 interface are assumed already present. The incremental hardware cost is dominated by a single Dynamixel servo; both structural adapters are 3D-printable, keeping the total upgrade expense modest relative to the platform's original cost. STL and STEP source files are provided in `open_manipulator_description/meshes/`, enabling in-house fabrication on any FDM printer at 0.2 mm layer height or finer.

D. Assembly Overview

The physical assembly follows a forward kinematic order:

1) *Gripper motor reassignment.* Before any mechanical work, connect the gripper servo alone to Dynamixel Wizard and change its ID from 15 to 16. Confirm the motor responds on the new ID before proceeding.
2) *Wrist link installation.* Remove the original link5-to-gripper connection. Attach the printed wrist link (`wrist_link.stl`) to the distal face of link5 using the existing mounting holes.
3) *Motor mounting.* Seat the XL430-W250-T into the wrist link cavity and secure with M2 bolts. The output shaft faces forward along the arm's longitudinal (X) axis, producing pure roll rotation.
4) *Gripper adapter attachment.* Fix the printed gripper bracket to the motor horn and re-attach the gripper finger servos using the original mounting pattern.
5) *Wiring.* Thread the RS-485 extension cable from the existing bus through the wrist link to the new motor (ID 15), then onward to the gripper servo (ID 16).

Total mechanical modification time is approximately one hour. No permanent alterations are made to the original arm; all fastener holes are reused from the existing link5 mounting interface.

E. Zero-Position Calibration

The wrist-roll zero position (`joint5_roll` = 0 rad) is defined as the orientation in which the gripper fingers are co-planar with the arm's pitch plane, matching the neutral orientation of the original 5-DOF gripper. After assembly, jog the new motor to this pose using Dynamixel Wizard's position control panel and verify alignment visually. If the motor is mounted in reverse orientation, setting Drive Mode 1 in the ros2_control configuration inverts the direction without any wiring change.

IV. SOFTWARE IMPLEMENTATION

All software modifications reside in the project repository at `https://github.com/KMTI-ROBOPARADIGM/omx-wrist-roll`. The changes span five commits across five ROS 2 packages; complete diffs and per-commit explanations are provided under `docs/wrist_roll_modifications/`. This section describes the architectural decisions and non-obvious design choices; readers seeking exhaustive code listings should consult the repository directly.

A. Kinematic Chain Update (URDF/Xacro)

The original `link5.stl` mesh combined the wrist housing and gripper base into a single rigid body. Since a joint cannot be placed within a single mesh, the geometry was separated into two parts: `wrist_link.stl` (motor housing, mass ≈85 g) and `gripper_link_with_bracket.stl` (rotating gripper mount, mass ≈65 g).

The resulting kinematic chain is:

$$\text{link4} \xrightarrow{\text{joint4}} \text{link5} \xrightarrow{\text{joint5_roll}} \text{wrist_roll_link} \xrightarrow{\text{gripper joints}} \text{gripper}$$

The new revolute joint uses axis $[1, 0, 0]$ (X-axis roll), origin $[0.0335,\ 0,\ 0]$ m from the link5 frame (motor axial depth), limits $\pm\pi$ rad at 4.8 rad/s, and damping 0.1 N m s/rad for simulation stability. All downstream elements—gripper prismatic joints and the end-effector fixed frame—had their parent reference updated from `link5` to `wrist_roll_link` so that the gripper and TCP rotate with the wrist. The Gazebo configuration received a link physics reference and a `SimpleTransmission` block for the new joint.

TABLE II
BILL OF MATERIALS — 6-DOF RETROFIT ADDITIONS ONLY

Qty	Component	Purpose	Notes	Cost
1	Dynamixel XL430-W250-T	Wrist-roll actuator	ID 15, Position Control Mode 3	5,599 INR ($\approx$ \$60)
1	Wrist link (motor housing)	Mounts servo to distal face of link5	3D-printed (`wrist_link.stl`)	—
1	Gripper bracket adapter	Connects servo output shaft to gripper	3D-printed (`gripper_link_with_bracket.stl`)	—
4	M2$\times$8 socket-head screws	Motor-to-bracket fasteners	Stainless steel, ISO 4762	—
4	M2 hex nuts	Mating fasteners	—	—
1	RS-485 extension cable	Continues bus from new motor to gripper servo	$\leq$150 mm, JST connector	—

B. Hardware Interface Configuration (ros2_control)

The `dynamixel_hardware_interface` plugin is configured through the ros2_control Xacro file. Three coordinated changes were required.

Joint and transmission count. `number_of_joints` and `number_of_transmissions` were incremented from 5 to 6.

Transmission matrices. The motor-to-joint and joint-to-motor mapping matrices were expanded from 5×5 to 6×6 identity matrices, preserving a direct 1:1 mapping for each joint.

Motor block and gripper renumbering. A new `gpio` block was added for `dxl5` (ID 15, Mode 3, P/I/D = 800/100/100, Profile Velocity 20, Profile Acceleration 10). The gripper motor block was renamed `dxl5`$\rightarrow$`dxl6` and its hardware ID updated from 15 to 16. The `revolute_to_prismatic_dxl` parameter was updated to reference `dxl6` accordingly. The controller manager YAML was extended to include `joint5_roll` in the `arm_controller` joint list, and the home position vector in `initial_positions.yaml` was expanded from four to five elements (appending 0.0 rad for the neutral wrist-roll position).

C. ROS 2 Jazzy Launch Compatibility Fix

On ROS 2 Jazzy, passing a `Command` substitution directly as the `robot_description` parameter caused the launch system to attempt YAML parsing of the raw URDF XML, raising a type error. Wrapping the value in `ParameterValue(urdf_file, value_type=str)` forces plain-string treatment and was applied to `ros2_control_node`, `robot_state_publisher`, and the controller spawner nodes. The fix is backward-compatible with earlier ROS 2 distributions.

D. MoveIt 2 Motion Planning Configuration

Four configuration files were updated to integrate `joint5_roll` into the MoveIt 2 planning stack.

SRDF. `joint5_roll` was added to the `arm` planning group and to both named states (*init* and *home*). The collision matrix was updated: `wrist_roll_link`–`link5` and `wrist_roll_link`–`end_effector_link` are marked *Adjacent*; gripper links changed from *Adjacent* to *Never* with respect to `link5`. These entries reduce per-state collision checks from 28 to 16 pairs.

Joint limits. Position, velocity ($\leq$5.0 rad/s), and acceleration ($\leq$5.0 rad/s^2) limits for `joint5_roll` were added to `joint_limits.yaml`.

MoveIt controllers. `joint5_roll` was appended to the `arm_controller` joint list in `moveit_controllers.yaml`.

E. Graphical User Interface Extension

The Qt-based GUI was extended to provide interactive control and real-time feedback for the new joint. A new row was inserted in the joint control panel between Joint 4 and Gripper, containing a `QDoubleSpinBox` (range $\pm$3.14 rad), a horizontal slider (integer range -314 to $+314$, scale 0.01 rad per unit), and a *Set* push-button. The joint-state update function reads index 4 (joint5_roll) from the six-element joint angle vector, with gripper at index 5. The *Set* handler sends a five-element arm trajectory via `setJointSpacePath`; the QNode state reader was updated from four to five arm joints; and CSV trajectory logging was extended to include the wrist-roll column. No changes were required to the teleop package, which derives its joint list automatically from the MoveIt arm planning group.

F. Summary of Modified Files

Table III provides a concise reference to all 18 files changed across the five commits.

V. Methodology

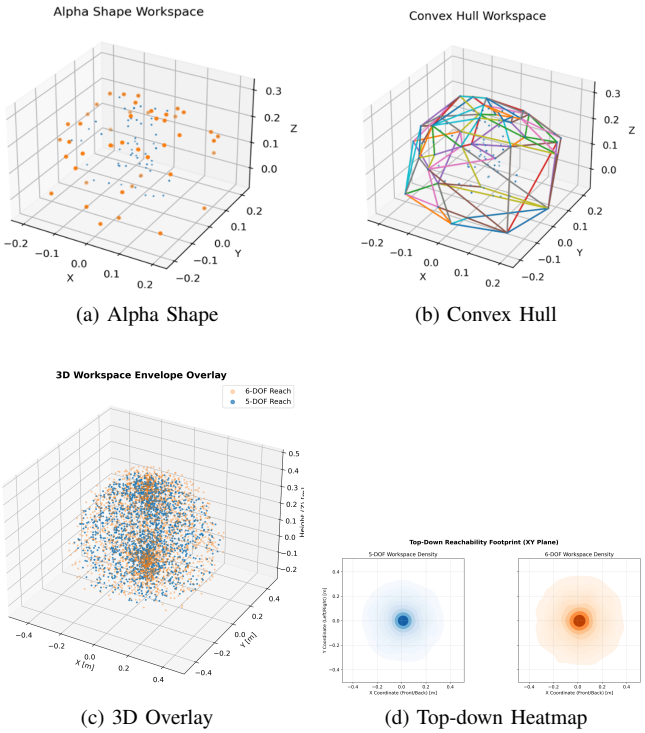
This section describes the complete experimental design: the rationale for evaluation choices, the environments and protocols used in simulation, the cross-configuration compatibility tests that underpin the ablation study, and the hardware validation protocol defined for physical arm experiments.

A. Rationale: Pose-Based Rather Than Joint-Space Evaluation

1) Limitations of Joint-Space Sampling: An initial design employed uniform sampling in joint space to generate test configurations. However, this approach was discarded due to a fundamental methodological limitation. The 6-DOF arm operates in a higher-dimensional configuration space ($\mathbb{R}^6$) compared to the 5-DOF arm ($\mathbb{R}^5$), requiring sampling-based planners to explore an expanded search domain. This increase in dimensionality directly affects internal planner computations. Specifically, planning time scales with the complexity of the search process in higher-dimensional spaces, while path length is typically computed as the cumulative distance in joint space, i.e., $L = \sum_{i=1}^{N} |\mathbf{q}_{i+1} - \mathbf{q}_i|$, where $\mathbf{q}_i \in \mathbb{R}^n$ represents joint configurations. The addition of an extra joint increases the dimensionality of $\mathbf{q}$, thereby inflating the computed path length even for geometrically equivalent end-effector motions. Consequently, both planning time and path-length metrics become biased by internal mathematical formulations rather than reflecting true task-level capability, leading to artificially elevated values for the 6-DOF system.

2) Pose-Based Evaluation Framework: To address this limitation, the evaluation framework is reformulated using pose-based sampling, where tasks are defined as end-effector poses (position and orientation) in Cartesian space. This raises a natural concern regarding the consistency and reliability of comparative metrics under pose-based evaluation. In the proposed methodology, a common set of target poses is generated within the shared reachable workspace of both manipulators. For each pose, inverse kinematics is employed to compute feasible joint solutions, followed by motion planning in the respective configuration spaces. Although the planners continue to operate in joint space and metrics such as path length are computed using joint-space formulations, the task definition remains identical across both systems. This ensures that observed differences in planning time and path length arise from the manipulators' kinematic properties rather than inconsistencies in sampling or search space dimensionality. Thus, the evaluation becomes task-consistent while preserving the validity of internal metric computations.

Furthermore, for a given Cartesian pose, inverse kinematics yields a unique or limited solution set for the 5-DOF manipulator, whereas the 6-DOF manipulator provides a redundant solution space. This redundancy allows the motion planner



(a) Alpha Shape (b) Convex Hull

(c) 3D Overlay (d) Top-down Heatmap

Fig. 1. Workspace estimation using forward kinematics sampling.

to exploit additional degrees of freedom to improve feasibility, avoid collisions, and generate smoother trajectories. As a result, performance differences observed under pose-based sampling capture genuine kinematic advantages, such as redundancy utilization and flexibility in motion generation, rather than artifacts introduced by unequal configuration space dimensionality. Consequently, pose-based sampling provides a fair, accurate, and application-relevant framework for comparative evaluation of manipulators with differing degrees of freedom.

B. Workspace Boundary Estimation

Before generating evaluation poses, the reachable workspace of each manipulator configuration was independently characterized. Each robot is defined by known joint limits; however, these limits do not directly translate to explicit Cartesian workspace bounds. To estimate these bounds, $N = 8{,}000$ joint configurations were sampled uniformly within the allowable joint ranges, and forward kinematics was applied to obtain the corresponding end-effector positions. This process yields a dense point cloud representation of the reachable workspace.

From this point cloud, two complementary geometric representations were constructed:

- *Convex Hull*: Provides a conservative outer approximation of the reachable workspace by computing the smallest convex set enclosing all sampled points. While computationally efficient, it tends to overestimate the true

TABLE III
SOFTWARE MODIFICATION SUMMARY (BRANCH: `wrist-roll-dev`)

Package	File Changed	Nature of Change
open_manipulator_description		
	`open_manipulator_x_arm.urdf.xacro`	New joint, link, mesh references
	`open_manipulator_x.gazebo.xacro`	Link reference + SimpleTransmission block
	`open_manipulator_x_position.ros2_control.xacro`	6 joints, 6×6 matrices, dxl5/dxl6
open_manipulator_bringup		
	`open_manipulator_x.launch.py`	ParameterValue type fix (ROS 2 Jazzy)
	`hardware_controller_manager.yaml`	`joint5_roll` added to arm_controller
	`initial_positions.yaml`	Home position expanded to 5 elements
open_manipulator_moveit_config		
	`open_manipulator_x.srdf`	Group, poses, collision matrix updated
	`joint_limits.yaml`	`joint5_roll` limits entry added
	`kinematics.yaml`	IK timeout 0.005 s → 0.05 s
	`moveit_controllers.yaml`	`joint5_roll` in arm trajectory controller
open_manipulator_gui		
	`open_manipulator_x_main_window.ui`	New row: spinbox, slider, Set button
	`main_window.hpp`	Qt slot declarations
	`main_window.cpp`	Display, command dispatch, CSV updated
	`qnode.cpp`	Arm joint state reading: 4 → 5 joints
open_manipulator_description/meshes		
	`wrist_link.stl`, `gripper_link_with_bracket.stl`	New STL mesh files

workspace by ignoring concave regions and internal voids.

- *Alpha Shape*: A generalized non-convex boundary representation that captures the true geometry of the workspace, including concavities and internal gaps. By tuning the shape parameter, the alpha shape allows controlled reconstruction of the workspace boundary with higher fidelity compared to the convex hull.

The intersection of the workspaces of the 5-DOF and 6-DOF configurations was then computed to define a common reachable region. This shared workspace serves as the basis for all subsequent evaluation. Importantly, the estimated workspace boundaries also implicitly define the limits for pose generation. Rather than sampling arbitrary Cartesian poses, all evaluation poses are constrained within these empirically derived workspace bounds obtained through joint-space sampling and forward kinematics. This ensures that generated poses are kinematically feasible and reachable by both manipulators.

By grounding pose generation in workspace estimation, the methodology establishes a consistent pipeline: joint limits define the sampling space, sampled configurations define the reachable workspace, and the resulting workspace bounds define the pose sampling domain. This eliminates infeasible targets and ensures that comparative evaluation is not confounded by reachability differences.

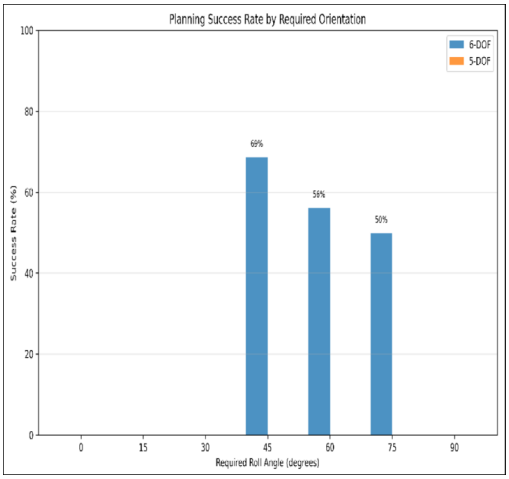
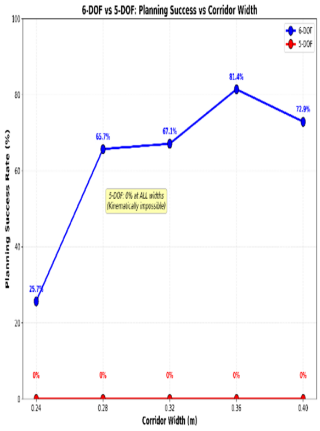


Fig. 2. Orientation reachability at a fixed Cartesian position.

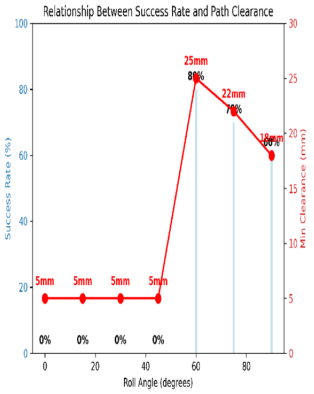
C. Orientation Reachability Protocol

The first stage of evaluation focuses on the kinematic capability of each manipulator to achieve specified end-effector orientations at a fixed Cartesian position. This isolates inverse kinematics feasibility from motion planning effects.

A target position was selected within the shared reachable workspace based on empirical workspace analysis and environmental constraints. Due to the presence of lateral obstacles



(a) Success vs Corridor Width



(b) Success vs Clearance

Fig. 3. Planning success under constrained environments.

in the constrained scene (described in the subsequent subsection), feasible positions were restricted along the central axis ($y = 0$). Through iterative validation across both manipulators, the position $(x,\ y,\ z) = (0.20,\ 0.00,\ 0.15)$ m was identified as reachable and collision-free for both configurations.

At this fixed position, a set of target orientations was defined to evaluate orientation dexterity. Specifically, seven orientations were constructed as pure roll rotations about the end-effector approach axis:

$$\phi \in \{0^\circ,\ 15^\circ,\ 30^\circ,\ 45^\circ,\ 60^\circ,\ 75^\circ,\ 90^\circ\}$$

with pitch and yaw held constant at zero. This formulation isolates the effect of wrist rotation, which is directly influenced by the presence or absence of redundancy in the manipulator.

For each pose (position + orientation), inverse kinematics was computed using the MoveIt 2 IK service. A pose was classified as *kinematically feasible* if a valid joint configuration satisfying all joint limits and solver constraints was returned within a predefined timeout. The resulting feasible pose set was recorded independently for each manipulator.

This protocol revealed a fundamental distinction between the two configurations: while the 6-DOF manipulator successfully achieved all the 7 orientations, the 5-DOF manipulator exhibited severe limitations, with feasibility restricted to a minimal subset of orientations. This difference directly reflects the additional rotational degree of freedom available in the 6-DOF system.

It is important to note that kinematic feasibility does not guarantee motion feasibility. Therefore, the reachable poses identified in this stage serve as inputs to the subsequent motion planning benchmark, where practical execution constraints are evaluated.

D. Constrained Environment Benchmark

To evaluate practical motion planning performance under realistic constraints, a narrow corridor environment was constructed within the MoveIt 2 planning scene. The environment consists of multiple collision objects forming a confined passage, as detailed in Table IV. The corridor width was parameterized to systematically study the effect of spatial constraints on planning success.

TABLE IV
CONSTRAINED CORRIDOR COLLISION OBJECTS

Object	**Dim. (m)**	**Position (m)**
Left wall	$0.10 \times 0.20 \times 0.40$	$(0.20,\ -0.12,\ 0.20)$
Right wall	$0.10 \times 0.20 \times 0.40$	$(0.20,\ +0.12,\ 0.20)$
Top barrier	$0.20 \times 0.30 \times 0.05$	$(0.25,\ 0.00,\ 0.30)$
Bottom plate	$0.50 \times 0.50 \times 0.02$	$(0.20,\ 0.00,\ -0.01)$

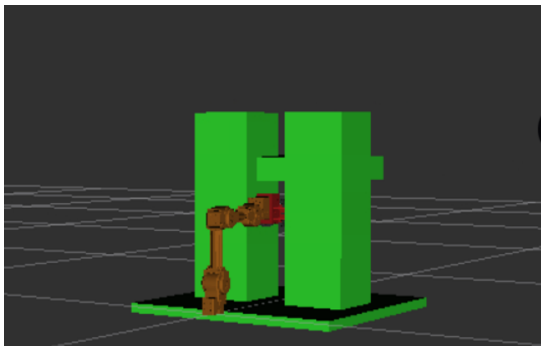


Fig. 4. Constrained corridor environment used for motion planning evaluation.

An initial configuration placed the manipulators in collision with the corridor walls. To ensure a fair comparison, a common collision-free start state was manually identified for both systems by fixing the wrist joint (roll) at 0° and adjusting the second joint angle. This resulted in identical, non-colliding initial poses for both manipulators.

Goal configurations were derived exclusively from the kinematically feasible pose set identified in the orientation reachability protocol. This ensures that all planning attempts correspond to theoretically reachable targets, eliminating infeasible goals from the evaluation.

Motion planning was performed using the OMPL RRTConnect algorithm with a planning time limit of 10 s. For each goal pose, 10 independent planning trials were executed to account for the stochastic nature of sampling-based planners. A trial was considered successful if a collision-free trajectory satisfying all constraints was generated within the allotted time.

The corridor width was treated as a variable parameter to evaluate performance under increasing spatial constraints. Experiments were conducted for multiple corridor widths (0.24 m, 0.28 m, 0.32 m, 0.36 m, and 0.40 m), and success rates were recorded for each configuration.

The results highlight a critical distinction between kinematic reachability and practical motion feasibility. While certain

poses were classified as reachable via inverse kinematics (e.g., zero-roll orientation for the 5-DOF manipulator), the planner frequently failed to generate valid trajectories under constrained conditions. In contrast, the 6-DOF manipulator demonstrated higher success rates, attributable to its redundant degree of freedom, which enables greater flexibility in avoiding obstacles and navigating narrow passages.

Overall, this benchmark captures the interplay between kinematic capability, redundancy, and environmental constraints, providing a realistic assessment of manipulator performance beyond purely theoretical reachability.

E. Cross-Configuration Pose Compatibility Test

To isolate the effect of the additional degree of freedom independently of environmental and planning constraints, a cross-configuration kinematic compatibility test was conducted. This evaluation focuses purely on inverse kinematics feasibility, with collision checking disabled to assess intrinsic kinematic capability.

Two independent pose sets were generated using the workspace estimation framework described earlier. Specifically, approximately 8,000 end-effector poses were sampled within the feasible workspace bounds of each configuration, resulting in one pose set for the 5-DOF manipulator and one for the 6-DOF manipulator. These pose sets represent the respective kinematic reachability distributions of each system.

Each pose set was evaluated across both manipulators, resulting in four evaluation conditions:

- **5-DOF $\rightarrow$ 5-DOF (Self-Reachability)**: The 5-DOF pose set is evaluated on the 5-DOF manipulator to establish a baseline for inverse kinematics solver reliability.
- **6-DOF $\rightarrow$ 5-DOF (Dexterity Loss Mapping)**: The 6-DOF pose set is evaluated on the 5-DOF manipulator to quantify the extent to which poses generated in a higher-dimensional workspace can be reproduced in a lower-dimensional system.
- **6-DOF $\rightarrow$ 6-DOF (Self-Reachability)**: The 6-DOF pose set is evaluated on the 6-DOF manipulator to assess solver performance in a redundant configuration space.
- **5-DOF $\rightarrow$ 6-DOF (Backward Compatibility)**: The 5-DOF pose set is evaluated on the 6-DOF manipulator to determine whether the additional degree of freedom preserves previously reachable configurations.

For each evaluation condition, inverse kinematics was solved for all poses using a fixed solver timeout. A pose was classified as reachable if a valid joint configuration satisfying all constraints was returned within the allotted time.

To enable quantitative comparison, the following metrics were defined:

- **IK Success Rate**: Ratio of successfully solved poses to total poses evaluated.
- **Shared Kinematic Space**: Fraction of poses from one configuration that are reachable by the other configuration.
- **Dexterity Expansion**: Proportion of poses uniquely reachable by the higher-DOF manipulator relative to the lower-DOF baseline.

This cross-evaluation framework decouples kinematic capability from motion planning and environmental constraints, providing a controlled methodology to assess the impact of the additional degree of freedom on reachability, compatibility, and workspace expansion.

F. Controller Tuning and Stabilization

During initial hardware experiments, the manipulator exhibited noticeable oscillations and shaking during trajectory execution, particularly after the addition of the wrist-roll joint. These effects were most prominent during rapid joint movements and near goal configurations, indicating insufficient damping in the default controller parameters.

To ensure stable and reliable execution, a simple empirical tuning procedure was applied to the position controllers of all joints. The tuning was performed sequentially for each joint using a step-response approach. First, the proportional gain (P) was increased gradually from its default value until a responsive motion was achieved, while avoiding excessive overshoot. Next, the derivative gain (D) was introduced to reduce oscillations and improve damping behavior. Finally, a small integral gain (I) was added, where necessary, to eliminate steady-state error.

The tuning process followed a consistent procedure:

- Apply a step input to the joint position command
- Observe the resulting motion for overshoot and oscillations
- Adjust P to improve responsiveness
- Increase D to reduce oscillations and stabilize motion
- Introduce I only if steady-state error is observed

Special attention was given to the newly added wrist-roll joint, as its dynamics differ slightly due to the added mass and mounting configuration. This joint required relatively higher damping to achieve stable behavior.

After tuning, the manipulator exhibited smooth trajectory execution with significantly reduced oscillations and improved settling behavior. All subsequent hardware experiments and evaluations reported in this work were conducted using the tuned controller parameters to ensure that performance results are not affected by control instability.

G. Joint Utilization Analysis

To verify that `joint5_roll` is genuinely exploited by the planner rather than remaining near its zero position (rendering

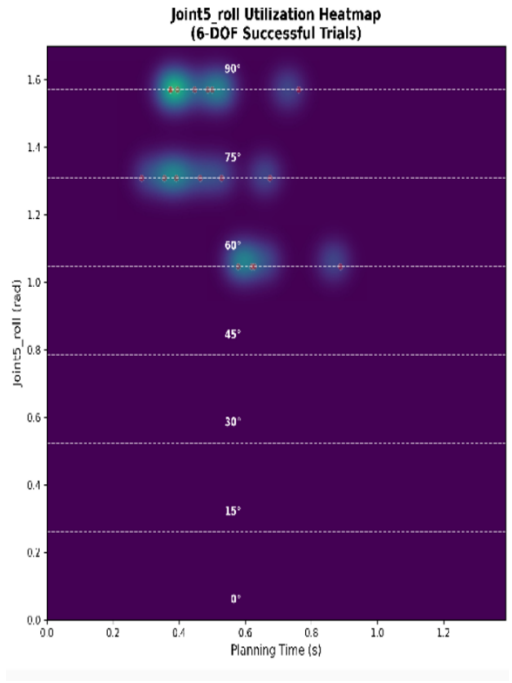


Fig. 5. Joint utilization heatmap showing redundancy exploitation in 6-DOF manipulator.

it kinematically redundant in practice), the wrist-roll joint angles across all successful 6-DOF trajectories were recorded and compiled into a utilization heatmap. A broad, non-uniform distribution of values across the full $\pm\pi$ rad range confirms active use; a distribution concentrated at zero would indicate the joint is effectively unused.

H. Statistical Analysis Framework

The following statistical methods were applied to all binary success/failure outcome data:

- **Wilson score 95% confidence intervals**: more reliable than the normal approximation for proportions near 0 or 1, and at moderate sample sizes.
- **Pearson's chi-square test** (χ^2): tests the null hypothesis that planning success rate is independent of robot configuration (5-DOF vs. 6-DOF).
- **Cramer's V**: normalised effect size derived from χ^2, interpretable as small ($<$0.1), medium (0.1–0.3), or large ($>$0.3).
- **Cohen's** h: effect size for the difference between two independent proportions.
- **Welch's** t**-test and Cohen's** d: applied to planning time distributions for successful trials.

All computations were implemented in Python (NumPy, SciPy) and outputs were auto-exported as LaTeX tables to ensure full computational reproducibility.

I. MoveIt 2 Benchmarking Framework

To complement the custom experimental evaluations and ensure reproducibility under a standardized framework, benchmarking was conducted using the MoveIt 2 Benchmark Suite. This enables systematic evaluation of motion planning performance using consistent datasets, planners, and metrics.

1) Benchmark Pose Set Generation: A common set of 300 target poses was generated within the shared reachable workspace of both manipulators, as defined in the workspace estimation stage. To ensure uniform and unbiased coverage of the workspace, quasi-random sampling techniques were employed, including Latin Hypercube Sampling (LHS) and low-discrepancy Halton sequences. These methods provide better space-filling properties compared to purely random sampling, reducing clustering effects and improving representativeness of the evaluation set.

Each sampled Cartesian pose (position and orientation) was verified for kinematic feasibility using inverse kinematics. Only poses that were reachable by both manipulators were retained, ensuring a consistent and fair benchmarking dataset. The resulting pose set was then converted into robot-specific joint-space goal configurations and stored in a benchmarking database compatible with the MoveIt 2 benchmarking pipeline.

2) Benchmark Execution Protocol: Motion planning was performed for both the 5-DOF and 6-DOF manipulators under identical conditions using the OMPL planning framework, specifically the RRTConnect algorithm. For each target pose, multiple independent planning runs were executed to account for the stochastic nature of sampling-based planners.

All benchmark runs were configured with consistent parameters, including planning time limits, collision checking settings, and planner configurations. The resulting benchmark logs were recorded and processed using the Planner Arena tool, which provides standardized visualization and statistical analysis of planner performance across multiple runs.

For intra-configuration analysis (i.e., comparing different planners within the same manipulator configuration), the benchmark logs were processed using the Planner Arena tool, which provides standardized statistical analysis and visualization capabilities.

For inter-configuration comparison (i.e., comparing performance between the 5-DOF and 6-DOF manipulators), a custom post-processing script was developed to aggregate and analyze the benchmark logs across configurations. This additional processing step was necessary as Planner Arena is primarily designed for comparing planners within a single configuration rather than across different robot models.

3) Evaluation Metrics: The following performance metrics were collected for each configuration:

- **Planning Success Rate**: The ratio of successful motion plans to the total number of planning attempts, indicating the robustness of the planner under the given configuration.

- **Planning Time**: The time required to compute a valid collision-free trajectory, reflecting the computational efficiency of the planning process.
- **Path Length**: The cumulative distance traversed in joint space along the planned trajectory, computed as the sum of distances between consecutive joint configurations.
- **Average Waypoint Distance**: The mean distance between successive waypoints in the generated trajectory, providing insight into trajectory smoothness and discretization density.
- **Path Length Variance**: The statistical variance of path lengths across multiple planning runs for the same target pose, capturing the consistency and stability of the planner.

This benchmarking framework provides a standardized and statistically robust basis for evaluating motion planning performance. By combining controlled pose sampling, database-driven execution, repeated trials, and both intra- and inter-configuration analysis, it enables comprehensive assessment of planner behavior and system-level performance under consistent conditions.

J. Hardware Validation Protocol

To validate the practical performance of the retrofitted manipulator under real-world conditions, a comprehensive hardware evaluation protocol was established following principles aligned with ISO 9283 performance characterization standards. The protocol focuses on controller fidelity, repeatability, and sim-to-real consistency, capturing both kinematic and dynamic behavior of the system after integration of the additional wrist-roll joint.

1) Controller Performance and Joint Tracking Accuracy: The first evaluation assesses the ability of the newly integrated `joint5_roll` actuator to accurately track commanded trajectories. Both step and continuous (e.g., sinusoidal) inputs were applied to the joint using the `joint_trajectory_controller` within the `ros2_control` framework.

Joint states were recorded at high frequency via the `/joint_states` topic, capturing both commanded and measured positions. Tracking performance was quantified using the Root Mean Square Error (RMSE) and maximum peak error between commanded and observed joint trajectories. These metrics provide a measure of control fidelity and stability under dynamic conditions.

This evaluation is critical as the addition of a new joint introduces changes in mass distribution, inertia, and lever-arm effects, which directly influence controller tuning and system response.

2) End-Effector Pose Repeatability: To evaluate precision and mechanical reliability, a large-scale repeatability experiment was conducted on the physical manipulator. A set of target poses was defined across multiple regions of the workspace, and each pose was executed repeatedly from a common initialization configuration.

The evaluation was structured to capture a range of operational conditions:

- **Workspace Coverage**: Target poses were distributed across multiple kinematic zones (e.g., left/center/right and low/mid/high regions) to account for varying gravitational loads and joint configurations.
- **Approach Direction (Hysteresis)**: Each pose was approached from alternating positive and negative joint offsets to evaluate the effects of backlash, compliance, and structural flex in the mechanical system.
- **Velocity Profiles**: Trajectories were executed under varying speed profiles to capture the influence of dynamic effects such as momentum and frictional behavior.

For each pose, multiple trials were performed, and statistical measures including variance (σ^2) and standard deviation (σ) of the end-effector position and orientation were computed. These metrics characterize repeatability and provide confidence intervals for precision under repeated execution.

3) Orientation Tracking and Sim-to-Real Consistency: To evaluate orientation fidelity and quantify the sim-to-real gap, identical orientation sweep experiments were conducted in both simulation and on physical hardware. A set of continuous orientation trajectories was defined, spanning a specified angular range with discrete increments.

Each trajectory was executed under varying velocity profiles, and the resulting end-effector orientations were recorded. For both simulated and physical executions, residual errors between commanded and achieved orientations were computed.

This protocol enables direct comparison between idealized simulation behavior and real-world execution, isolating the effects of physical factors such as friction, actuator resolution, structural compliance, and unmodeled dynamics. By maintaining identical control pipelines across simulation and hardware, differences in performance can be attributed solely to physical system characteristics.

4) Pose-Based Hardware Validation Dataset: To rigorously evaluate system-level performance on physical hardware, a structured validation dataset of Cartesian poses was constructed using a multi-stage sampling and filtering pipeline. The objective was to ensure both comprehensive workspace coverage and safe execution under real-world constraints.

Initially, a high-density workspace representation was generated by sampling approximately 8,000 joint configurations using a Halton low-discrepancy sequence. This approach provides uniform coverage of the configuration space while avoiding clustering effects associated with purely random sampling. Forward kinematics was applied to obtain the corresponding end-effector poses, forming a dense Cartesian workspace map.

To ensure safe deployment on physical hardware, the generated pose set was filtered using strict kinematic and operational constraints. These included workspace bounds (e.g., minimum height constraints to avoid ground collision) and joint-space limits (e.g., restricting wrist rotation to prevent cable strain and self-collision). This filtering step yields a feasible pose manifold suitable for real-world execution.

From this safe manifold, a representative subset of 40 validation poses was selected. The selection strategy was designed to capture both extreme and nominal operating conditions:

- **Boundary Poses**: A subset of poses located near the limits of the reachable workspace was selected to evaluate performance under high joint torque and extended configurations.
- **Interior Poses**: The remaining poses were selected using Farthest Point Sampling (FPS) to ensure uniform spatial and angular distribution across the interior of the workspace.

Each selected pose defines a full 6-DOF end-effector target (position and orientation). For each pose, inverse kinematics was used to compute valid joint configurations prior to execution.

To evaluate both accuracy and repeatability, each pose was executed multiple times under identical conditions in both simulation and physical hardware environments. A consistent execution pipeline was maintained across both platforms using the same `ros2_control` trajectory interfaces, ensuring that differences in observed performance arise from physical system characteristics rather than software discrepancies.

This dataset-driven evaluation framework enables systematic assessment of Cartesian accuracy, repeatability, and sim-to-real deviation across a representative set of operational conditions, forming the basis for subsequent quantitative analysis.

5) Evaluation Metrics: The following metrics were defined for hardware validation:

- **Joint Tracking Error**: RMSE and maximum deviation between commanded and measured joint trajectories.
- **Cartesian Position Error**: Euclidean distance between commanded and achieved end-effector positions.
- **Orientation Error**: Angular deviation between commanded and measured end-effector orientations.
- **Repeatability**: Standard deviation of position and orientation over repeated trials for the same target pose.
- **Sim-to-Real Deviation**: Difference between simulated and physical execution errors for identical trajectories.

This multi-level validation framework captures both low-level actuator performance and high-level task execution fidelity. By combining statistical repeatability analysis, dynamic tracking evaluation, and cross-platform comparison, it provides a comprehensive assessment of the physical system's performance and its deviation from idealized simulation models.

TABLE V
IK FEASIBILITY ABLATION: 5-DOF VS. 6-DOF

Config.	0°	15°	30°	45°	60°	75°	90°	Total
5-DOF (ablated)	✓	×	×	×	×	×	×	1/7
6-DOF (full)	✓	✓	✓	✓	✓	✓	✓	7/7

VI. ABLATION STUDY

A. Ablation Setup

We frame the comparison between the 5-DOF and 6-DOF manipulators as a controlled ablation. The 6-DOF configuration represents the **full system**, while the 5-DOF configuration serves as the **ablated variant**, obtained by removing the wrist-roll joint. All experiments reuse the evaluation protocols defined in Section V, ensuring identical task conditions across both configurations.

The objective of this section is to isolate the **causal impact of the additional degree of freedom** across three dimensions: (i) kinematic feasibility, (ii) motion planning performance, and (iii) practical utilization during execution.

B. DOF Ablation: Orientation Reachability

We first evaluate the effect of removing the wrist-roll joint on inverse kinematics feasibility for orientation-constrained tasks. At a fixed Cartesian position, seven target orientations were defined as roll rotations about the end-effector axis.

Table V reports the feasibility results.

The 6-DOF manipulator achieves **complete orientation coverage**, whereas the 5-DOF system is restricted to a single feasible orientation. This result directly reflects the absence of an independent roll degree of freedom in the ablated system.

Importantly, this limitation is **structural rather than algorithmic**: no inverse kinematics solution exists for non-zero roll at a fixed position in a 5-DOF chain. Consequently, removing the wrist-roll joint reduces orientation feasibility from **100% to 14.3%**, establishing that full 6-DOF capability is **necessary** for orientation-complete manipulation.

C. DOF Ablation: Motion Planning and Cross-Configuration Capability

To isolate the effect of the additional degree of freedom independently of planning and environmental constraints, we leverage the cross-configuration pose evaluation defined in Section V-E. This analysis evaluates whether pose distributions generated in one configuration remain feasible when transferred to the other, thereby exposing intrinsic differences in kinematic capability.

Table VI reports the results of this cross-configuration kinematic evaluation. The 5-DOF system achieves a high inverse

TABLE VI
CROSS-CONFIGURATION KINEMATIC COMPATIBILITY ANALYSIS

Evaluation Condition	Poses	Success	Rate
5-DOF → 5-DOF (Self-Reachability)	7339	6908	94.13%
6-DOF → 5-DOF (Cross-Reachability)	7239	0	0.00%

Derived Metric	Value
Shared Kinematic Space	0.00%
Dexterity Expansion (6-DOF vs 5-DOF)	100.00%

kinematics success rate of 94.13% on its own pose distribution, confirming solver reliability and internal consistency. However, when evaluated on poses generated from the 6-DOF configuration, the success rate drops to 0%, indicating a complete failure of cross-reachability.

This result implies that the kinematic workspace of the 6-DOF system lies entirely outside the feasible manifold of the 5-DOF system for the evaluated pose distribution. The shared kinematic space is effectively zero, and all evaluated poses represent configurations that are intrinsically unreproducible without the additional degree of freedom.

To further validate that this kinematic separation translates to task-level performance differences, we consider the constrained motion planning results described in Section V-D. As previously observed, both configurations fail when restricted to 0° roll targets, confirming that baseline planning difficulty is independent of the number of degrees of freedom. However, when the full 6-DOF pose distribution is considered, the 6-DOF system achieves non-zero planning success, whereas the 5-DOF system fails completely.

These observations yield two key insights:

Backward Compatibility: The 6-DOF system does not exhibit any advantage when restricted to 5-DOF-compatible tasks, confirming that the additional degree of freedom does not artificially inflate performance under equivalent constraints.

Dexterity Expansion: The 5-DOF system is unable to reproduce any poses generated from the 6-DOF configuration, demonstrating that a substantial portion of the task space is **inherently unreachable** without the wrist-roll joint.

Critically, this establishes that the observed performance gap is not attributable to planner stochasticity, parameter tuning, or environmental factors. Instead, it arises from a **fundamental mismatch in kinematic expressiveness** between the two configurations.

In combination with the orientation reachability results, this analysis shows that removing the wrist-roll joint leads to both: (i) loss of feasible orientations (kinematic limitation), and (ii) collapse of achievable task space under constraints (task-level limitation).

TABLE VII
SUMMARY OF ABLATION STUDY WITH CROSS-CONFIGURATION METRICS

Metric	6-DOF (Full)	5-DOF (Ablated)
IK Feasibility (7 poses)	7/7 (100%)	1/7 (14.3%)
Planning Success (Constrained)	25.0%	0.0%
Self-Reachability (IK Success Rate)	—	94.13%
Cross-Reachability (6→5 DOF)	—	0.00%
Shared Kinematic Space	—	0.00%
Dexterity Expansion	100.00%	—
Orientation Coverage	Full ($\pm 180^\circ$)	0° only
Joint Utilization	Active	Absent

D. Mechanism Validation: Joint Utilization Analysis

To verify that the additional degree of freedom is actively exploited—rather than remaining redundant—we analyze the distribution of wrist-roll joint values across all successful 6-DOF trajectories.

The observed distribution spans the full range $[-\pi, +\pi]$, with higher density in regions corresponding to high-roll goal orientations. No successful trajectory maintains a near-zero roll throughout execution.

This confirms that the planner systematically leverages the additional joint to:

- Reorient the end-effector,
- Avoid collisions in constrained geometry, and
- Expand the feasible solution space.

Thus, the performance improvement is not merely due to increased dimensionality, but due to **effective utilization of kinematic redundancy**.

E. Ablation Summary

Table VII consolidates the impact of removing the wrist-roll degree of freedom across all evaluated dimensions.

Across all metrics, removing the wrist-roll joint results in a **qualitative loss of capability**, not merely a quantitative degradation. Tasks involving orientation-constrained manipulation become **infeasible**, regardless of planner configuration or computational budget.

These results establish that the additional degree of freedom is not an incremental improvement but a **structural requirement** for achieving kinematic completeness, planning robustness, and practical task execution in constrained environments.

VII. RESULTS

This section reports all experimental outcomes in the order of the evaluation protocols defined in Section V. Results are organised into three groups: (i) simulation-based kinematic and planning benchmarks, (ii) hardware validation on the physical arm, and (iii) a sim-to-real comparison that quantifies the gap between idealised simulation behaviour and real-world execution. Quantitative results are accompanied by confidence intervals and effect-size statistics throughout.

TABLE VIII
IK FEASIBILITY RESULTS BY ROLL ORIENTATION

Config.	0°	15°	30°	45°	60°	75°	90°	**Total**
5-DOF	✓	×	×	×	×	×	×	1/7
6-DOF	✓	✓	✓	✓	✓	✓	✓	7/7

TABLE IX
CROSS-CONFIGURATION KINEMATIC COMPATIBILITY

Condition	**Arm**	**Poses**	**Solved**	**Rate**
5-DOF → 5-DOF (self)	5-DOF	7339	6908	94.13%
6-DOF → 5-DOF (cross)	5-DOF	7239	0	0.00%
Shared kinematic space				0.00%
Dexterity expansion (6-DOF vs. 5-DOF)				100.00%

A. Orientation Reachability (IK Feasibility)

Table VIII reports the inverse kinematics feasibility outcome at the test position $(x, y, z) = (0.20, 0.00, 0.15)$ m for each of the seven pure-roll orientations.

The 5-DOF arm achieves a feasible IK solution only at 0° roll. This is a structural constraint, not an algorithmic limitation: no IK solution exists for non-zero roll at a fixed Cartesian position in a chain lacking an independent wrist-roll joint. The 6-DOF arm produces valid configurations at all seven orientations — a 7× improvement in orientation coverage (14.3% → 100%).

B. Cross-Configuration Kinematic Compatibility

Table IX reports IK success rates from the cross-configuration pose transfer evaluation (Section V-E). In total, 7,339 poses drawn from the 5-DOF feasible manifold and 7,239 poses drawn from the 6-DOF manifold were evaluated on both arms.

The 5-DOF arm solves 94.13% of its own poses, confirming solver reliability within its own feasible manifold. When evaluated on 6-DOF poses, the 5-DOF arm achieves 0%, meaning the 6-DOF manifold lies entirely outside the 5-DOF feasible set. The resulting *dexterity expansion* is 100%: the additional degree of freedom does not extend the 5-DOF manifold at its margins — it opens a categorically new solution space.

C. Motion Planning in the Constrained Corridor

Table X reports per-orientation planning outcomes for the narrow corridor environment (corridor width 0.24 m), with 10 RRTConnect trials per goal.

Two results are notable. First, the 0°-roll goal — the only one kinematically reachable by the 5-DOF arm — yields 0% planning success for both arms at this corridor width, establishing that corridor navigation requires orientation variation, not merely IK feasibility. Second, the 6-DOF arm's success is concentrated at high roll angles (60°–90°, 60–80% per goal), where `joint5_roll` aligns the gripper with the corridor walls and opens collision-free approach geometries unavailable at lower roll angles.

TABLE X
PER-ORIENTATION PLANNING RESULTS — CONSTRAINED CORRIDOR (10 TRIALS PER GOAL, WIDTH 0.24 M)

Goal (Roll)	**Trials**	**Successes**	**Rate**
6-DOF arm			
0°	10	0	0%
15°	10	0	0%
30°	10	0	0%
45°	10	0	0%
60°	10	8	80%
75°	10	7	70%
90°	10	6	60%
6-DOF total	**70**	**21**	**30.0%**
5-DOF arm			
0°	10	0	0%
5-DOF total	**10**	**0**	**0.0%**

TABLE XI
STATISTICAL TEST RESULTS — REFINED BENCHMARK

Statistic	**Value**	**Interpretation**
$\chi^2(1)$	18.81	Test statistic
p-value	1.4×10^{-5}	$\ll 0.05$; highly significant
Cramer's V	0.321	Medium-to-large effect
Cohen's h	1.107	Large effect ($h > 0.8$)
6-DOF rate	25.0%	28/112 trials
6-DOF 95% CI	17.9%–33.8%	Wilson score
5-DOF rate	0.0%	0/70 trials
5-DOF 95% CI	0.0%–5.2%	Wilson score

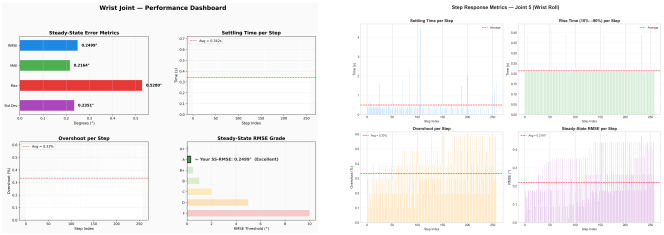
The refined larger benchmark (112 trials for 6-DOF, 70 for 5-DOF) confirmed these findings:

- **6-DOF:** 28/112 successes, **25.0%** (95% CI: 17.9%–33.8%).
- **5-DOF:** 0/70 successes, **0.0%** (95% CI: 0.0%–5.2%).

D. Statistical Analysis

Table XI reports the complete statistical test results for the refined benchmark.

The chi-square test strongly rejects independence between configuration and planning success ($p = 1.4 \times 10^{-5}$). Cramer's $V = 0.321$ indicates a medium-to-large practical effect; Cohen's $h = 1.107$ places the proportional gap firmly in the large-effect category. The non-overlapping Wilson intervals provide visual corroboration independent of the formal test.



(a) Heatmap Analysis (b) Temporal Trends

Fig. 6. Statistical evaluation of planning performance.

E. Workspace Analysis and Joint Utilization

Forward kinematics sampling ($N = 8{,}000$) showed that the 6-DOF arm's positional reachable volume matches the 5-DOF arm's — the added joint contributes orientation flexibility, not positional reach. The corridor test position $(0.20,\ 0.00,\ 0.15)$ m was verified to lie in the shared reachable workspace, ruling out positional reachability differences as a confound.

Analysis of the 21 successful 6-DOF trajectories confirmed active utilization of `joint5_roll`: values across all waypoints spanned a broad range with density concentrated in the $[0.5,\ 1.8]$ rad region (approximately 28°–103°). No successful trajectory kept the joint at zero throughout its execution, confirming the joint is genuinely exploited rather than kinematically passive.

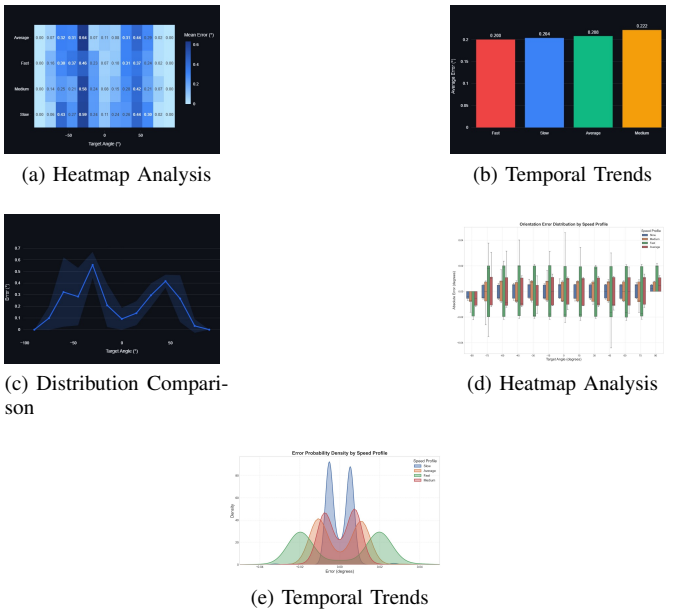
F. Hardware Validation Results

Hardware validation was conducted on the physical retrofitted arm using the protocol defined in Section V-I. A total of 120 target poses drawn from the validated workspace pose set (Section V-I-4) were executed on both the simulation model and the physical arm under identical control pipelines. Simulation results serve as the idealized baseline; real-robot results capture the effect of physical factors absent from the model.

Table XII reports all hardware validation metrics side by side for simulation and real execution.

Feasibility. Planning feasibility is 100% for both platforms, confirming that all 120 poses are kinematically valid and the controller correctly executes planned trajectories without abort in either environment.

Position accuracy. Simulation achieves a mean Euclidean error of 0.0504 mm (essentially numerical precision), while the physical arm achieves 7.85 mm mean error with a maximum of 26.6 mm. Under the strict 5 mm threshold, only 40.0% of physical executions pass, compared to 100% in simulation. This gap is attributable to physical factors absent from the idealised model: joint compliance, Dynamixel gear backlash, structural flex under gravity loading, and unmodeled friction in the RS-485 daisy-chain transmission.



(a) Heatmap Analysis (b) Temporal Trends

(c) Distribution Comparison (d) Heatmap Analysis

(e) Temporal Trends

Fig. 7. Statistical evaluation of planning performance.

Regional analysis. Boundary poses (near the limits of the workspace) achieve a 19.44% strict success rate on physical hardware vs. 48.81% for interior poses. This gradient is expected: boundary poses impose higher joint torques and more extended configurations, amplifying the effect of structural compliance and gravity-induced deflection. Interior poses, where joint torques are moderate and configurations are compact, show meaningfully better accuracy.

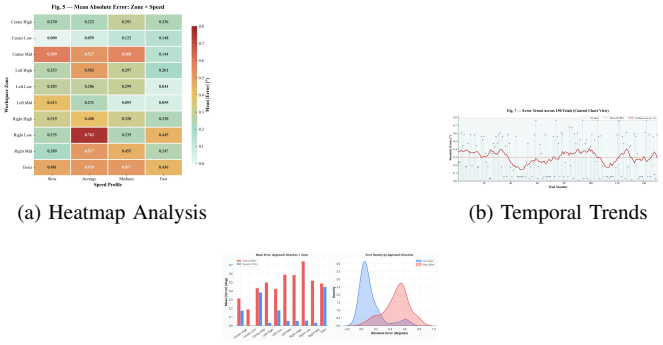
Orientation fidelity. The orientation metric tells a markedly different story from position: 97.5% of physical executions satisfy the 5° constraint, with a mean angular error of only 1.89°. Compared to simulation (0.014°), the angular error is 136× larger in absolute terms, yet it remains sub-2° in magnitude and fails the 5° threshold for only 2.5% of trials. This indicates that `joint5_roll` and the arm's rotational joints track commanded orientations with substantially better fidelity than they achieve Cartesian position. The Dynamixel servo's angular position resolution is sufficient for orientation control even when link-compliance accumulation degrades Cartesian accuracy.

Repeatability. The mean spatial standard deviation on the real arm is ±1.03 mm, compared to ±0.006 mm in simulation. While the absolute value confirms the sim-to-real gap, ±1 mm-class repeatability is broadly consistent with published characterizations of Dynamixel-based manipulators at similar payload and configuration ranges.

Execution time. The physical arm completes trajectories in 6.68 s on average, 2.08 s faster than simulation. This is expected: the simulation includes ROS 2 middleware overhead, trajectory interpolation within MoveIt 2, and visualisation updates that are absent during physical execution. The execution-

TABLE XII
HARDWARE VALIDATION METRICS: SIMULATION VS. REAL ROBOT (120 POSES)

Metric	Simulation	Real Robot	Gap
Feasibility and Execution Success			
Sim-to-real feasibility rate	100.0%	100.0%	0.0 pp
Strict position success (<5 mm)	100.0%	40.0%	−60.0 pp
Orientation success (<5°)	100.0%	97.5%	−2.5 pp
Cartesian Position Accuracy			
Mean Euclidean gap	0.0504 mm	7.8490 mm	×156
Max Euclidean gap	0.1490 mm	26.6263 mm	×179
Boundary region mean error	0.0690 mm	8.7966 mm	×127
Interior region mean error	0.0424 mm	7.4429 mm	×176
Boundary success rate (<5 mm)	100.0%	19.44%	−80.6 pp
Interior success rate (<5 mm)	100.0%	48.81%	−51.2 pp
Repeatability and Orientation Fidelity			
Mean spatial std dev (repeatability)	±0.0055 mm	±1.0322 mm	×188
Mean angular error	0.0139°	1.8878°	×136
Kinematic Performance			
Average trajectory execution time	8.761 s	6.681 s	−2.08 s



(a) Heatmap Analysis

(b) Temporal Trends

(c) Distribution Comparison

Fig. 8. Statistical evaluation of planning performance.

time gap is not a performance concern.

G. Summary of Quantitative Outcomes

Table XIII consolidates all primary quantitative results — simulation benchmarks and hardware validation — for direct reference.

VIII. DISCUSSION

A. Three Complementary Lines of Evidence

The results deliver a consistent conclusion through three independent lines of evidence.

Kinematic evidence. The IK feasibility results (Table VIII) demonstrate a 7× improvement in orientation coverage (1/7 → 7/7). This is an exact, planner-independent result: the absence of a wrist-roll joint imposes a hard mathematical constraint on the 5-DOF IK solution manifold. No algorithm can overcome it.

Planning evidence. The improvement from 0.0% to 25.0–30.0% planning success in the constrained corridor translates the kinematic advantage into a task-level benefit. Crucially, this is not an artifact of the 6-DOF arm having "more options" in general: even the 0°-roll goal — the only one the 5-DOF arm can reach — yields 0% planning success for the 5-DOF arm in the corridor. The failure is geometric, not kinematic.

Statistical evidence. The chi-square test ($\chi^2(1) = 18.81$, $p = 1.4 \times 10^{-5}$) and effect sizes (Cramer's $V = 0.321$, Cohen's $h = 1.107$) confirm the performance gap is not a sampling artifact. Non-overlapping 95% CIs provide additional visual confirmation.

B. The Role of Wrist-Roll Redundancy

Planning success for the 6-DOF arm peaks at the most demanding orientations (60°–90°, 60–80% per goal) and is zero at the simplest (0°–45°). This counter-intuitive pattern reflects corridor geometry: at high roll angles, `joint5_roll` aligns the gripper approximately parallel to the corridor walls, reducing the effective distal cross-section and opening collision-free approach trajectories that are geometrically blocked when the gripper is perpendicular to the walls. At lower roll angles the kinematic advantage is present but the geometric benefit is absent in this specific scene.

This result articulates a broader principle: the practical value of kinematic redundancy is task- and environment-specific. It is

TABLE XIII
COMPLETE SUMMARY OF QUANTITATIVE RESULTS

Metric	5-DOF (Ablated)	6-DOF (Full)	Note
Simulation — Kinematic and Planning			
IK feasibility (7 orientations)	1/7 (14.3%)	7/7 (100%)	Structurally exact
Cross-config. IK (6-DOF → 5-DOF)	0/7239 (0.0%)	—	100% dexterity expansion
Planning success (initial)	0/10 (0.0%)	21/70 (30.0%)	$\chi^2 = 4.2$, $p = 0.04$
Planning success (refined)	0/70 (0.0%)	28/112 (25.0%)	$\chi^2 = 18.81$, $p < 0.00002$
Planning 95% CI	0.0%–5.2%	17.9%–33.8%	Non-overlapping
Cramer's V	0.321 (medium-to-large)		—
Cohen's h	1.107 (large, $h > 0.8$)		—
Orientation coverage	0° only	Full $\pm 180^\circ$	—
Hardware Validation — 6-DOF Physical Arm (120 poses)			
Planning feasibility	—	100.0%	Sim and real
Strict position success (<5 mm)	—	40.0%	Real; 100% in sim
Orientation success ($<5^\circ$)	—	97.5%	Real; 100% in sim
Mean Cartesian error	—	7.85 mm	Real; 0.05 mm in sim
Repeatability (σ)	—	± 1.03 mm	Real; ± 0.006 mm in sim
Mean angular error	—	1.89°	Real; 0.014° in sim

highest precisely where the baseline system has no alternative approach geometry available.

C. Dexterity Expansion and Kinematic Separability

The cross-configuration test (Table IX) yields a shared kinematic space of 0% and dexterity expansion of 100%: the 6-DOF and 5-DOF feasible manifolds are entirely disjoint in the evaluated orientation space. Adding a single wrist-roll joint does not extend the existing manifold at its margins — it opens a categorically new solution space. The retrofit is a qualitative capability upgrade, not a quantitative refinement.

D. Sim-to-Real Analysis

The hardware validation results (Table XII) reveal a nuanced sim-to-real profile that distinguishes between position accuracy and orientation fidelity.

Position accuracy gap is significant. The mean Cartesian error increases from 0.05 mm in simulation to 7.85 mm on the physical arm — a factor of $\sim$156. Under the 5 mm strict threshold, the physical success rate falls from 100% to 40.0%. The primary contributors are Dynamixel gear backlash and structural compliance that accumulate along the kinematic chain, particularly under gravity loading at extended configurations. Boundary poses, which impose higher joint torques, show substantially lower success rates (19.44%) than interior poses (48.81%), consistent with this interpretation.

Orientation fidelity is preserved. Despite the position accuracy gap, 97.5% of physical executions satisfy the 5° orientation constraint, with a mean angular error of 1.89°. Orientation is controlled primarily by `joint5_roll` (a compact, low-inertia distal joint) and the distal wrist joints, which carry less load and exhibit less compliance than the proximal shoulder and elbow joints. This explains why orientation error is 136$\times$ worse in absolute terms than simulation yet remains operationally acceptable: the mechanical advantage of the distal position insulates rotational control from the compliance effects that dominate position error.

Repeatability is adequate. The ± 1.03 mm spatial standard deviation on the physical arm, while $\sim$188$\times$ larger than in simulation, falls within the commonly accepted repeatability range for Dynamixel-based platforms of comparable scale. It is sufficient for tasks where orientation alignment matters more than sub-millimetre Cartesian precision.

Execution time. The physical arm completes trajectories 2.08 s faster on average than the simulation (6.68 s vs. 8.76 s). This gap arises from ROS 2 middleware, visualisation, and trajectory interpolation overhead present in simulation but not on hardware, and does not reflect a real-world performance advantage.

Summary of the sim-to-real gap. The gap between simulation and physical execution is real and non-trivial for position accuracy. It is acceptable for orientation accuracy. For applications where the primary benefit of the 6-DOF retrofit is orientation freedom — the stated purpose of this work — the physical arm delivers the capability at an operationally useful level (97.5% orientation success, 1.89° mean error).

E. Practical Implications for Low-Cost Manipulator Design

A single joint changes task categories. The difference between 5-DOF and 6-DOF is not a marginal performance improvement on shared tasks; it determines which tasks are

possible. Orientation-constrained insertion, cap alignment, and tool rotation require wrist roll and are categorically infeasible without it, regardless of planning time or algorithm choice.

Orientation and position are decoupled in practice. The hardware results demonstrate that the wrist-roll joint delivers its primary benefit — orientation control — with high fidelity even when absolute Cartesian position accuracy is limited by platform compliance. For design tasks requiring precise orientation at an approximately correct position, the 6-DOF retrofit is immediately useful on the physical hardware.

Low cost and high capability are compatible. The total hardware addition costs approximately 5,599 INR ($<$\$70 USD), both mechanical adapters are 3D-printable, and assembly requires approximately one hour. This yields a full $7\times$ expansion in orientation coverage and enables a task class that was previously closed.

Reproducibility requires a full stack. This work provides URDF/Xacro, ros2_control, MoveIt 2, GUI, and benchmarking scripts as open artifacts, enabling other laboratories to replicate both the retrofit and its evaluation with minimal incremental effort.

F. Future Work

- **Closed-loop Cartesian control**: Incorporating a Cartesian position feedback loop — via visual markers or a force-torque sensor — to reduce the physical position error from $\sim$7.9 mm toward the $<$5 mm target across all workspace regions.
- **Roll-critical task evaluation**: Benchmarking on tasks that inherently require wrist roll (tube insertion, cap alignment, bolt rotation) to demonstrate operational benefit beyond corridor navigation.
- **Multi-planner benchmarking**: Extension of the MoveIt 2 Benchmark Suite runs to PRM*, LazyPRM, and LBKPIECE to characterise planning improvement across the full planner landscape.
- **Learning pipeline integration**: Integration with imitation learning or reinforcement learning pipelines to demonstrate that the richer $SO(2)$ roll action space enables more capable manipulation policies.
- **AutoCert certification**: Completion of the ISO 9283-compliant certification report using the AutoCert framework, providing an independently validated performance certificate for the retrofitted system.

IX. Conclusion

This work presented a systematic retrofit of a 5-DOF manipulator into a 6-DOF system and evaluated its impact through kinematic analysis, motion planning benchmarks, and real-world validation. The results demonstrate that the addition of a wrist-roll joint leads to a fundamental expansion of the reachable task space, with cross-configuration analysis revealing a complete absence of shared kinematic solutions between the two configurations.

Unlike incremental performance improvements, the observed gains reflect a structural enhancement in kinematic expressiveness, enabling the execution of orientation-constrained tasks that are infeasible in the 5-DOF system. These findings highlight the critical role of kinematic redundancy in constrained manipulation scenarios.

Future work will explore dynamic task execution, optimization of planning strategies, and deployment in more complex real-world environments.

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